

Gavin White

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Skills

Programming Languages: Python, C#, Java, C++, C, JavaScript, HTML, CSS

Development Tools: Git, Docker, Kubernetes, .NET, Kafka, Jenkins, Hangfire, Datadog, GraphQL, REST, Google Cloud Platform, MSSQL, PostgreSQL, Elasticsearch, Node.js

Experience

Software Engineer II, Wayfair – Boston, MA

Mar 2026 – May 2026

Software Engineer, Wayfair – Boston, MA

Aug 2024 – Mar 2026

- Developed low-latency REST APIs to access and update order data, supporting over 200 million daily requests
- Replatformed backend order placement infrastructure with an asynchronous, decoupled architecture and a sharded database for multi-datacenter support, improving fault tolerance and resilience to dependency outages
- Built a concurrent task-processing framework for order placement, enabling orchestration of interdependent workflows across downstream domains to improve data consistency and processing reliability
- Engineered an extensible custom metadata system for order and order product entities, enabling zero-code field onboarding, reduced query overhead through optional loading, automatic Elasticsearch index mapping updates, and rapid delivery of new high-revenue purchasing channels
- Developed a reusable outbox-based event service to produce platform events for 20+ event types, improving reliability and observability of event delivery while enabling migration from unstable legacy infrastructure

Software Engineer Co-op, Delsys – Natick, MA

Aug 2023 – Dec 2023

- Engineered multi-modal sensor systems for human motion analysis and non-contact physiological monitoring, synchronizing data from RGB, thermal, infrared, and depth cameras
- Developed real-time calibration software for three-dimensional, arbitrary-perspective camera arrays using checkerboard calibration and ICP point cloud registration
- Optimized calibration with C++ multi-threading and CUDA, resulting in an 87.5% reduction in calibration time
- Researched and applied computer vision and deep learning methods in Python to extract physiological signals from video; refactored the software pipeline for modularity and future extensibility

Software Engineer Co-op, Wayfair – Boston, MA

Jul 2022 – Dec 2022

- Built decoupled Python microservices to process and transform high-volume financial data, enabling reliable tracking and storage of business-critical information
- Designed database schemas, implemented GraphQL queries, and developed Kafka-based event pipelines to support scalable, real-time financial data processing
- Implemented observability tooling with Datadog and Grafana to detect process failures and data anomalies

Projects

Connect Four Robot

- Ideated, constructed, and programmed a robot that plays Connect Four automatically using an Arduino to drive motors that move the robot and a camera to capture board state in real time
- Developed a computer vision program to determine board state from an image and artificial intelligence minimax algorithm to choose the optimal move for the robot to play

Wordle Heuristics Engine

- Designed and implemented an algorithm to help users play Wordle more efficiently using statistical heuristics
- Implemented MVC architecture and command pattern for an extensible design with interchangeable strategies

Education

Northeastern University – Bachelor of Science in Computer Science

May 2024

GPA: 3.9/4.0, *Summa Cum Laude*

Relevant Coursework: Software Engineering, Object-Oriented Design, Algorithms & Data, Computer Systems, Web Development, Database Design, Compiler Design, Programming Languages, Networks & Distributed Systems